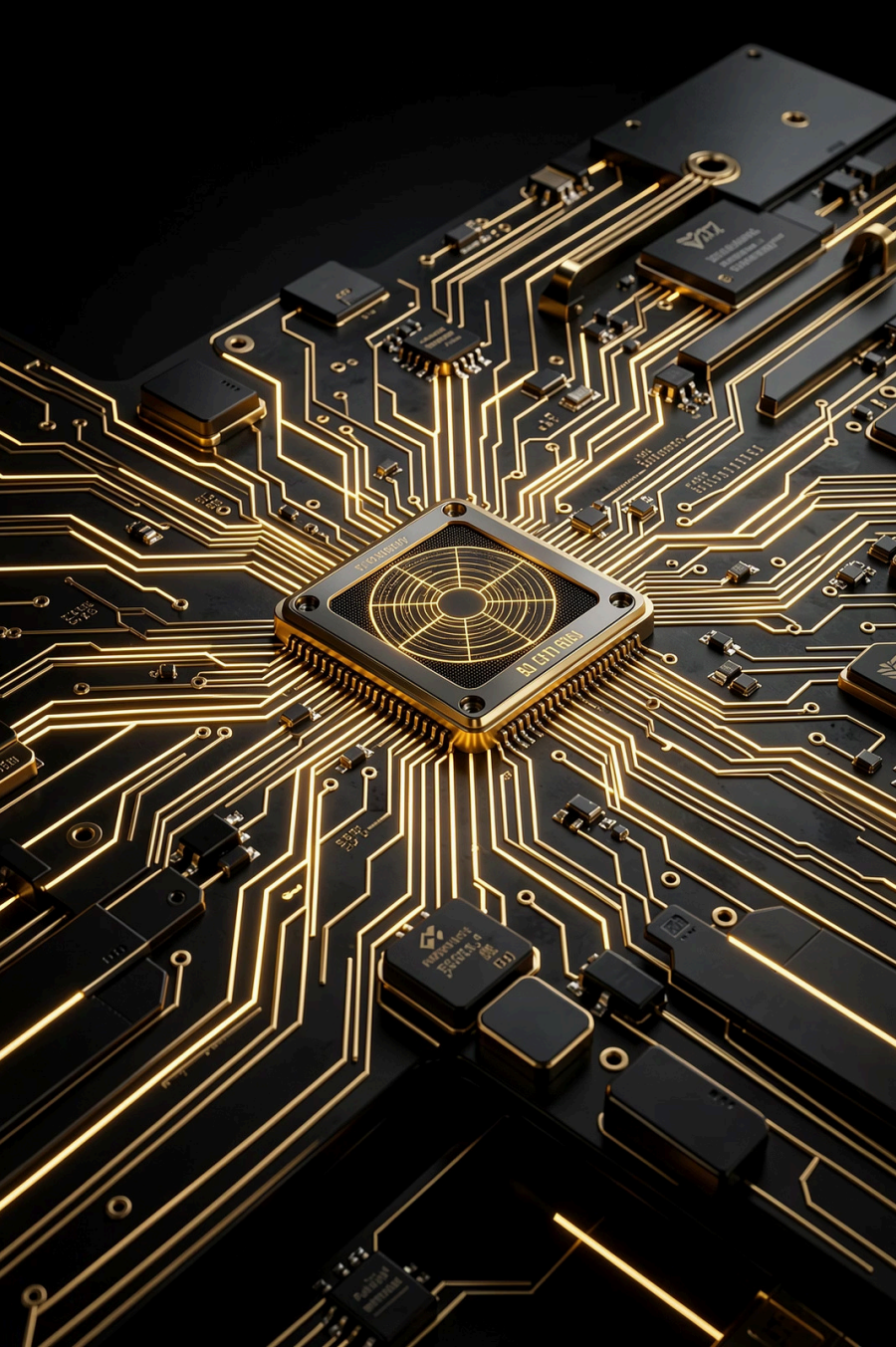


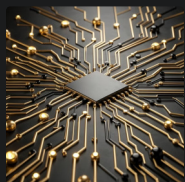


Hardware Trojan Detection

AI-Powered RTL Security Analysis

Machine Learning Framework for Detecting Trojans
in Verilog RTL Code

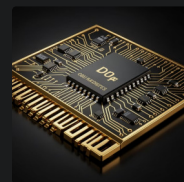
Dataset Overview



14 TJ-RTL-toy
RTL Level Trojans



74 IP-Netlist-toy
Netlist Trojans



21 IP-RTL-toy
Simple IP Trojans



3 TEST_FILES
Test Samples

- PIC16F84, RS232 variants
- ISCAS C432, C499, C880
- Adder, BCD, Encoder
- SRAM, UART trojans

Total: 112 Hardware Trojans + 11 Clean Reference Designs

Types of Hardware Trojans

Trigger Mechanisms

Counter-based

Activates after N clock cycles, often hiding until a specific event or time threshold is met.

State Machine

Detects a specific sequence of inputs or internal states (e.g., AA, 55, 22, FF) before activation.

Comparator

Triggers when specific signal conditions or data values are met, such as a particular input combination.

Time-based

Activates after a predefined duration or an event like a 32-bit overflow, making detection difficult during normal operation.

Payload Effects

Denial of Service

Causes signals to become stuck at 0/1, rendering the hardware inoperable or severely degrading performance.

Information Leakage

Performs bit replacement or stealthily exfiltrates sensitive data from the system, compromising confidentiality.

Reliability Reduction

Introduces intermittent failures, increasing error rates, or shortening the hardware's lifespan without immediate detection.

Functional Change

Modifies the intended circuit behavior, leading to incorrect computations, altered logic, or unauthorized operations.

Feature Extraction (62 Features)

Structural

- Line/character counts
- Module count
- Input/output ports
- Wire/register signals
- Always blocks
- Control flow (if/case statements)

Operators

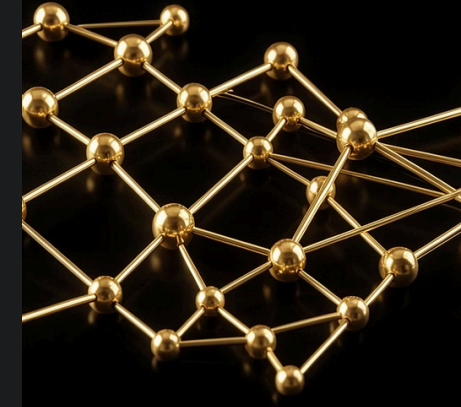
- Logical (&&, ||, !)
- Bitwise (&, |, ^)
- Comparison (==, !=)
- Arithmetic (+, -, *)
- Shift (<<, >>)
- Ternary (?:)

Suspicious

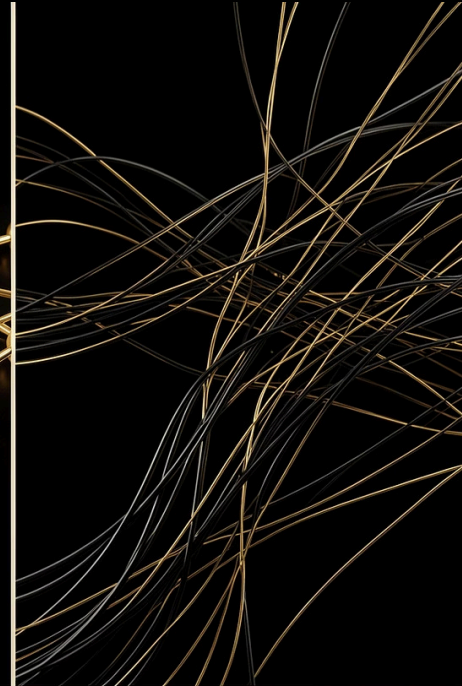
- Magic constants
- Counter patterns
- State machines
- Comparison chains
- Stuck-at patterns
- Force statements

These 62 features provide a comprehensive profile of the RTL code, enabling the AI model to identify both known and novel hardware Trojan characteristics. The combination of structural, operational, and suspicious patterns helps in creating a robust detection framework.

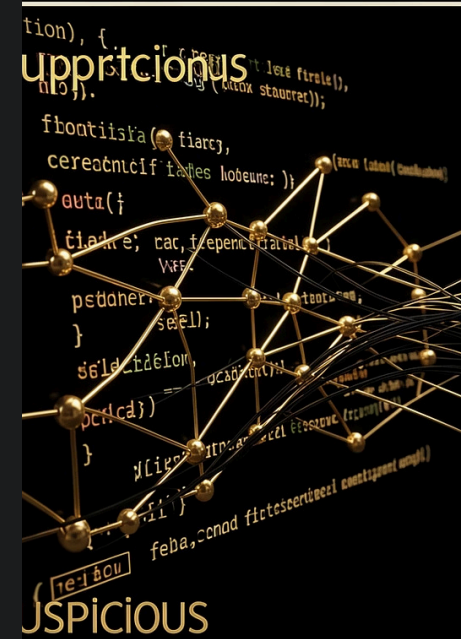
STRUCTURAL



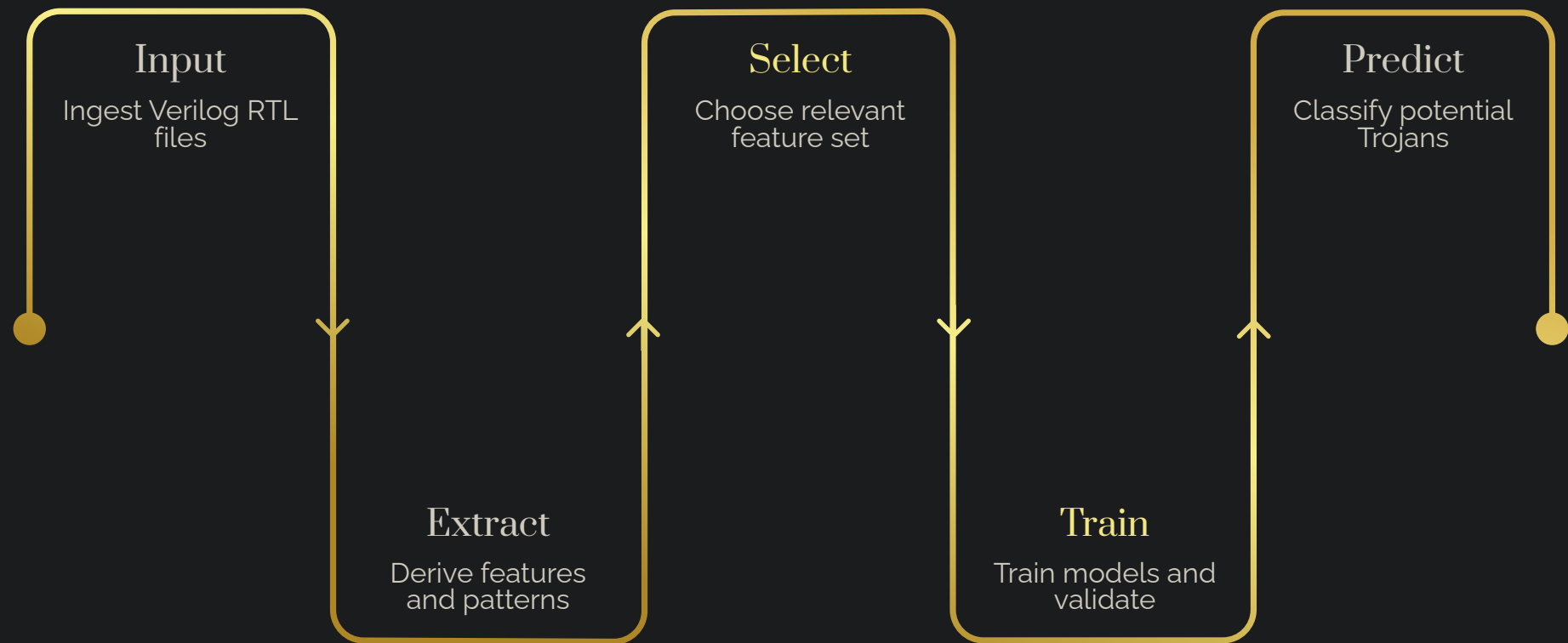
OPERATIONAL



SUSPICIOUS



AI Model Pipeline



The AI model pipeline meticulously processes Verilog RTL code through a series of stages to identify potential hardware Trojans. This systematic approach ensures thorough analysis and accurate classification.

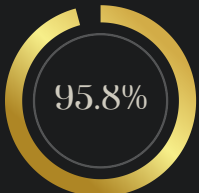
8 Models Compared:

KNN	MLP
AdaBoost	Gradient Boosting
Random Forest	Decision Tree
Logistic Regression	SVM

Model Performance



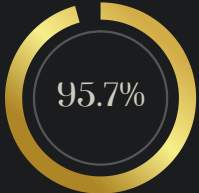
Accuracy



F1 Score



Recall



ROC-AUC

Confusion Matrix

Predicted		Clean		Trojan	
Actual	Clean	0	2		
	Trojan	0	23		

The model demonstrates excellent performance, particularly with 100% recall for Trojan detection, meaning no actual Trojans were missed. This is crucial for hardware security where false negatives can have severe consequences.

Top Selected Features

- 1. total_chars
- 2. output_count
- 3. always_blocks
- 4. case_statements
- 5. posedge_count

These features proved most indicative in distinguishing between clean and Trojaned RTL code, highlighting the model's ability to pinpoint critical structural and behavioral anomalies.



Top Selected Security Analysis

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- Cldh6 j233%
- Ussis jinenadesl

How to Use

Predict Trojans

```
from predict import TrojanPredictor

predictor = TrojanPredictor(
    'models/ht_detector.pkl'
)
result = predictor.predict_file(
    'design.v'
)
```

Easily integrate the pre-trained model into your workflow to quickly scan Verilog files for potential hardware Trojans. The predictor provides a straightforward interface for immediate security analysis.

Train New Model

```
from train_model import \
    train_and_save_best_model

results = train_and_save_best_model(
    './your_dataset',
    './models'
)
```

Customize the detection framework by training new models with your own datasets. This flexibility allows for adaptation to specific design environments and emerging Trojan types.

Files Included

- `feature_extractor.py` — Extract 62 features from Verilog
- `train_model.py` — Train and compare ML models
- `predict.py` — Make predictions on new files
- `deep_learning_model.py` — CNN/LSTM/Transformer models for advanced analysis
- `models/ht_detector_model.pkl` — Pre-trained model (ready to use!)

Key Takeaways

AI can effectively detect hardware trojans in RTL code

Leveraging machine learning provides an automated and scalable solution to a critical security challenge.

92% accuracy with 100% recall on trojan detection

Ensuring that no malicious insertions go unnoticed, offering robust protection.

Identifies suspicious patterns: counters, state machines, magic constants

The framework excels at pinpointing specific indicators of Trojan presence.

Multiple model options: Traditional ML and Deep Learning

Providing flexibility and advanced capabilities for diverse detection scenarios.

Ready-to-use framework with pre-trained models

Enabling immediate deployment and rapid assessment of RTL code security.

Start Detecting Trojans Today!